Introduction to Git and GitHub
Part 1

Kendra Oudyk (she/her)

Many parts of this presentation are inspired / based on these great resources

- Chacon, S., & Straub, B. (2014). *Pro git.* Springer Nature. Available at https://git-scm.com/book/en/v2
- The Carpentries. (2021). *Version Control with Git.* https://swcarpentry.github.io/git-novice/.

Land Acknowledgement

University of British Columbia, Point Grey Campus (**Vancouver**)

We would like to begin by acknowledging that the land on which we gather is the traditional, ancestral, and unceded territory of the **xwməθkwəy̓əm** (Musqueam) People.

Find out about the land where you live at https://native-land.ca/

Who am I?

- Multiple fields of study / research
  - BMus → MA → PhD (Neuroscience) → MLIS (Library Studies)

- Graduate Academic Assistant in Digital Scholarship

Who are you?

1. I study/research…

2. I write code…

   - Never / a few times
   - Monthly
   - Weekly
   - Daily

Side note on learning strategies

- **Dual coding**
  - Hence the visualizations alongside the terminal
  - Draw diagrams

- **Active retrieval**
  - Try to remember before looking something up

- **Immediate feedback**
  - Make low-stakes mistakes
  - Volunteer incorrect answers here

- **Spaced practice**
  - Keep practicing (e.g., put code for courses on GitHub)
  - Attend workshops again

These slides complement the **self-paced** materials on the Research Commons website

https://ubc-library-rc.github.io/intro-git/

Goals

Part 1

- What is distributed version control?
- Why is Git useful?
- Track your own work with Git

Part 2 (next week)

- Share your work on GitHub (simple / linear collaboration)
- Collaborate through GitHub (complex / nonlinear collaboration)
- Navigating GitHub
- Try it out!

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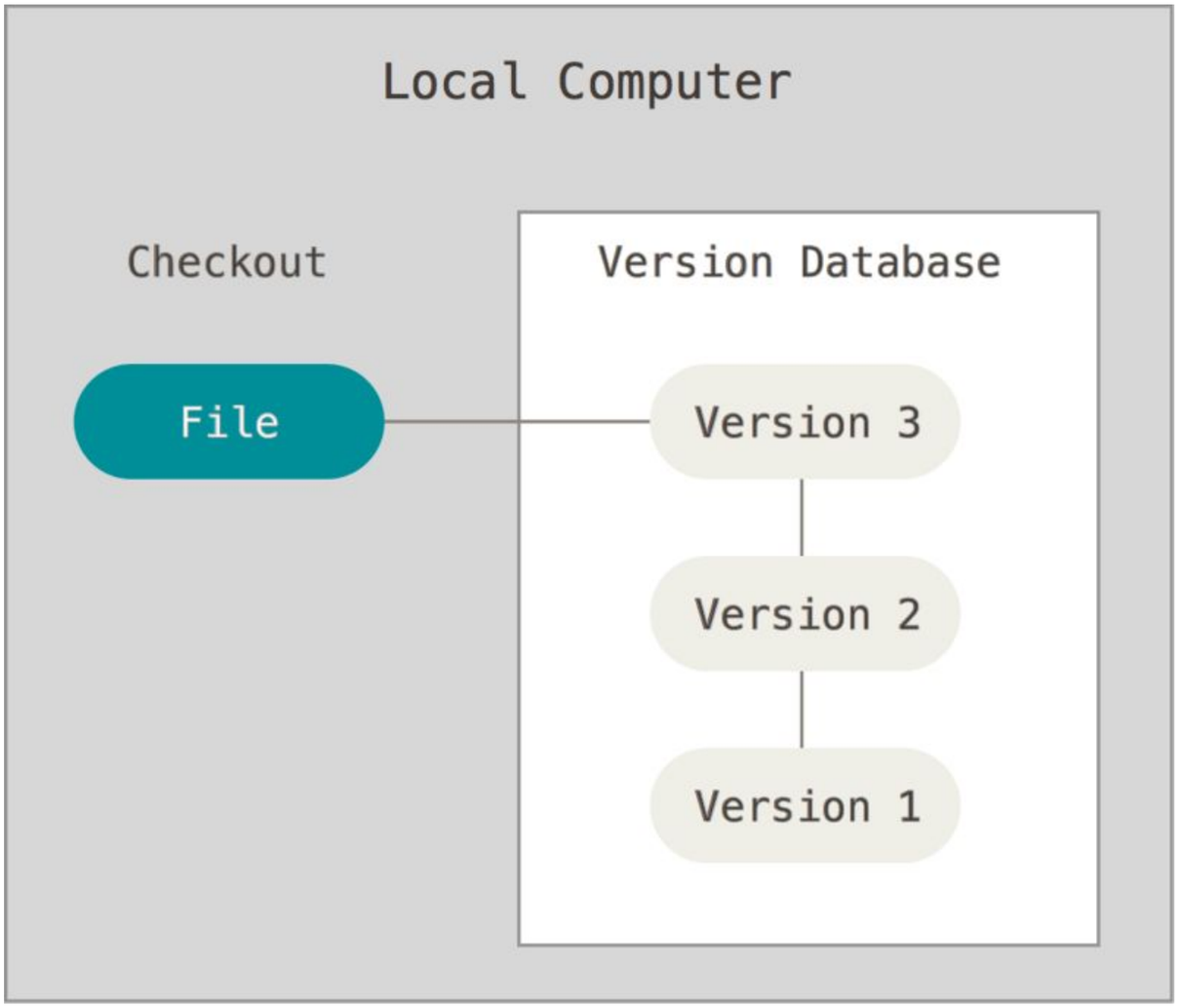
Version control

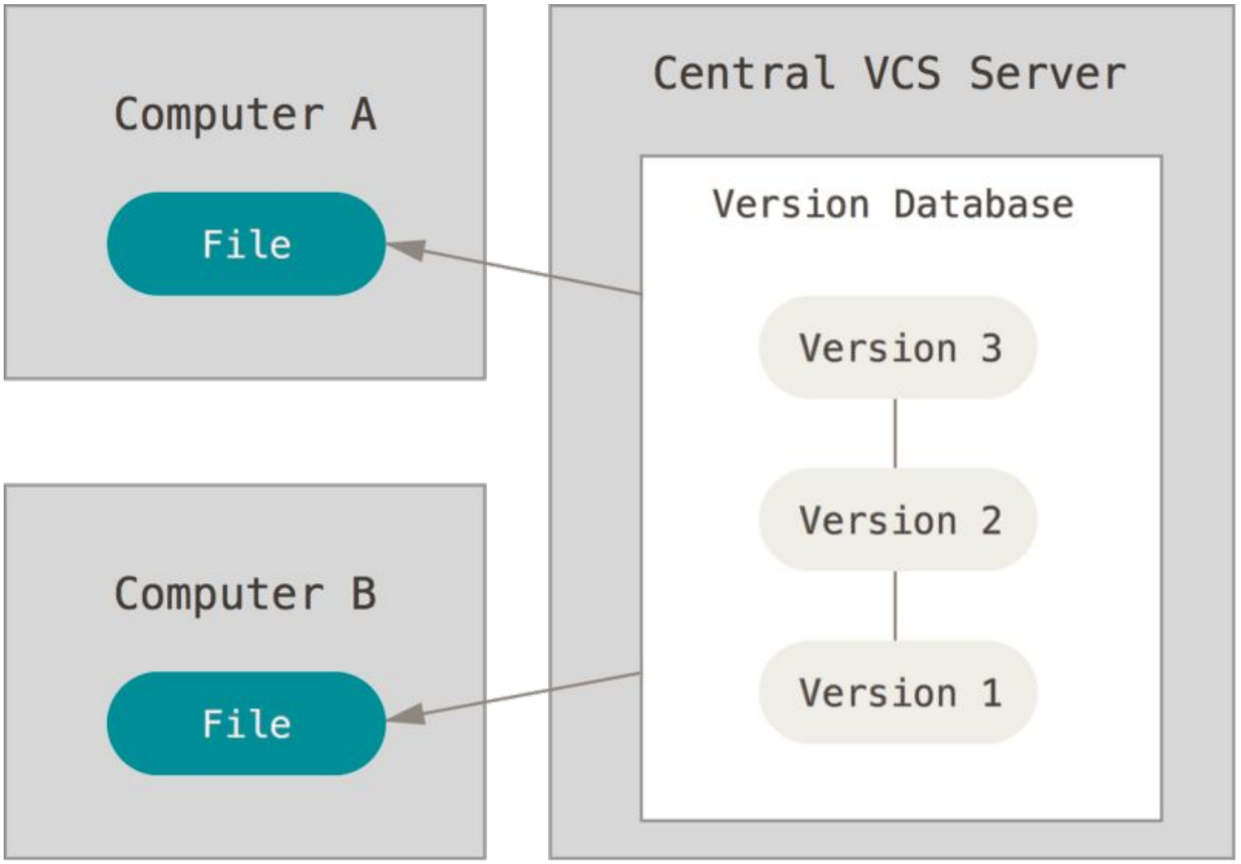
- Tracks **changes** to files
- Lets you recall different **versions** of files
- Becomes more essential as **collaborative projects** grow
- Can track almost **any type** of file (works best on text-based files)
- There are different **types** of version control

Local version control



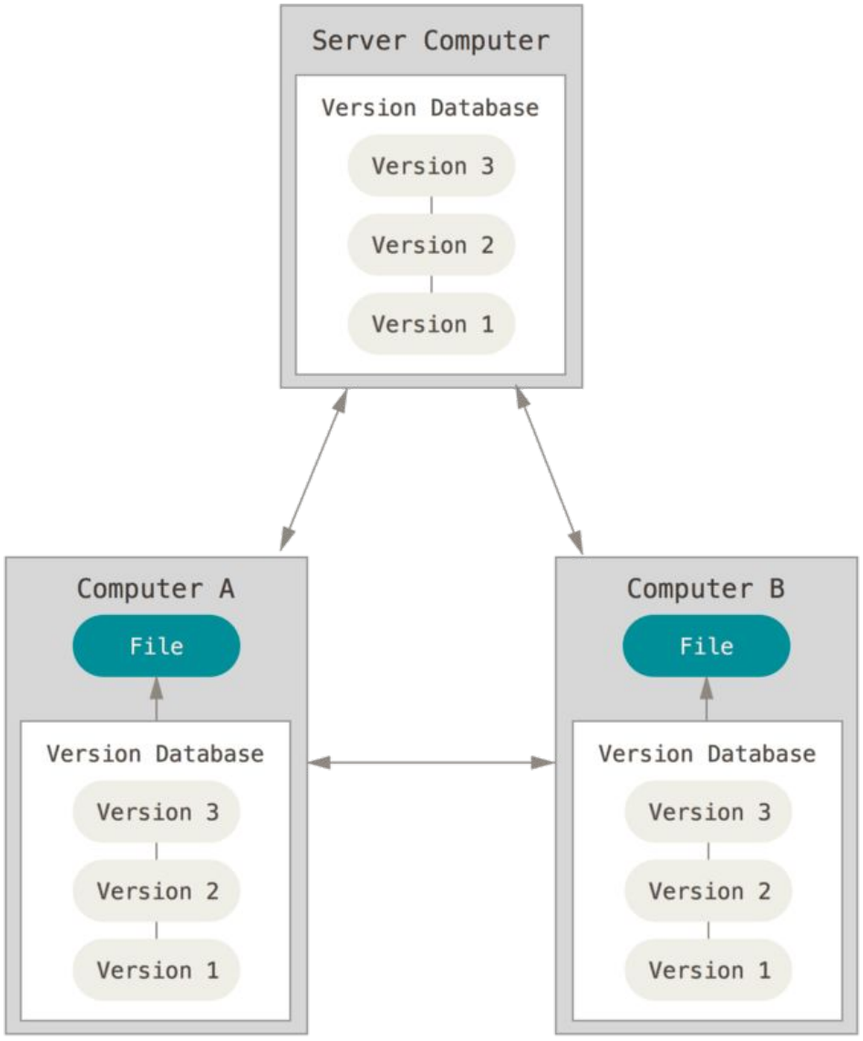
But how do we collaborate?

Centralized version control



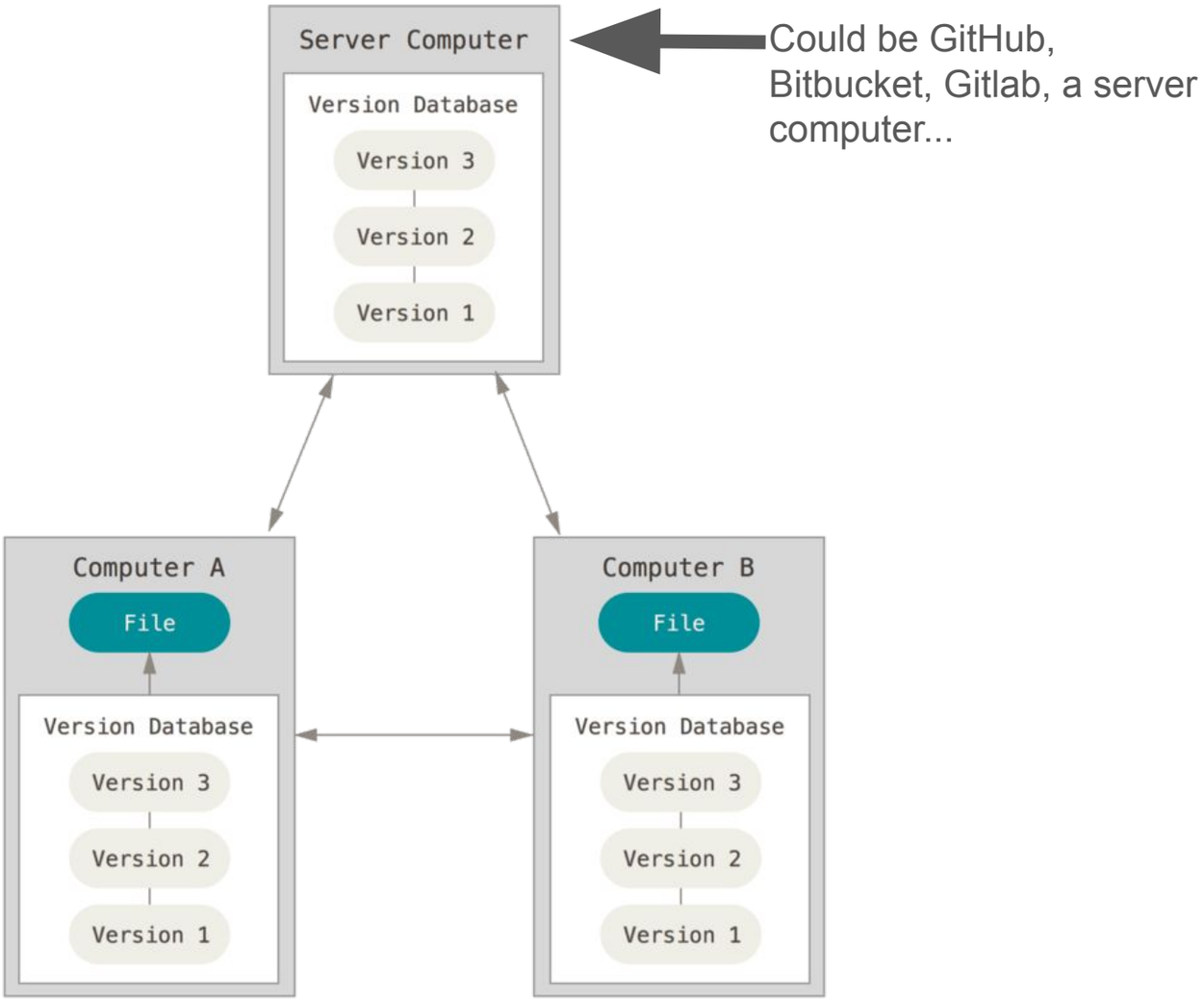
But what if the server crashes?

Distributed version control



Git vs. GitHub

- Git is the "**language**" we use to do version control.
- GitHub **hosts** git repositories online.



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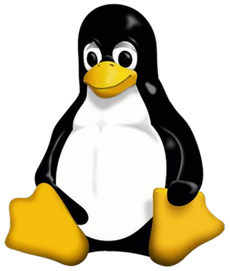
git was made for Linux



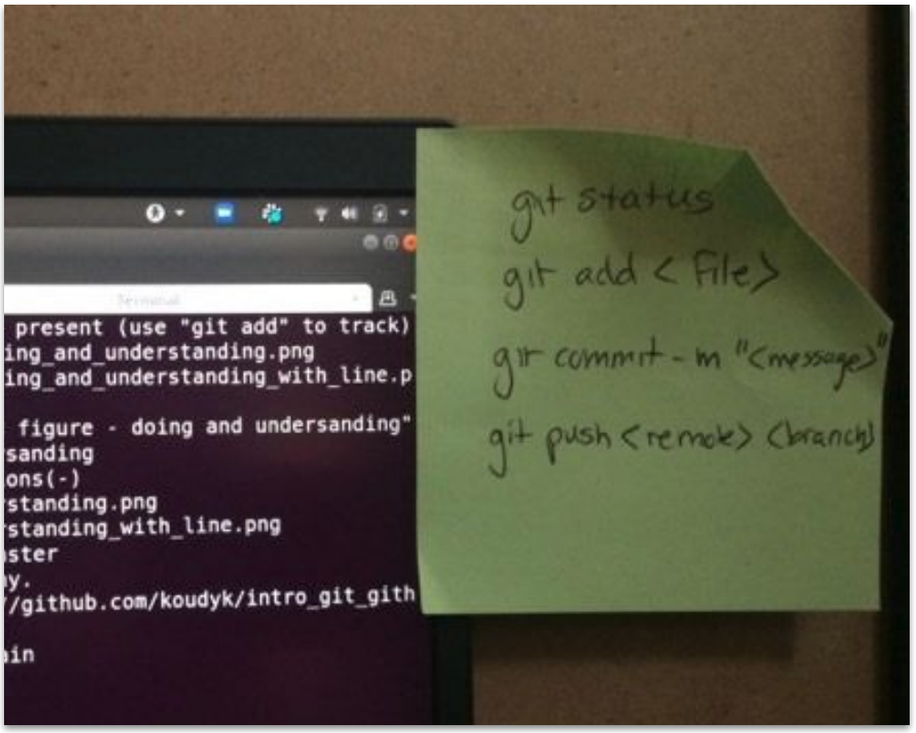
i.e., a huge open-source project.

- In the last month, “1067 authors have pushed 5,329 commits…” [https://github.com/torvalds/linux/pulse/monthly, as of 2021-07-14]

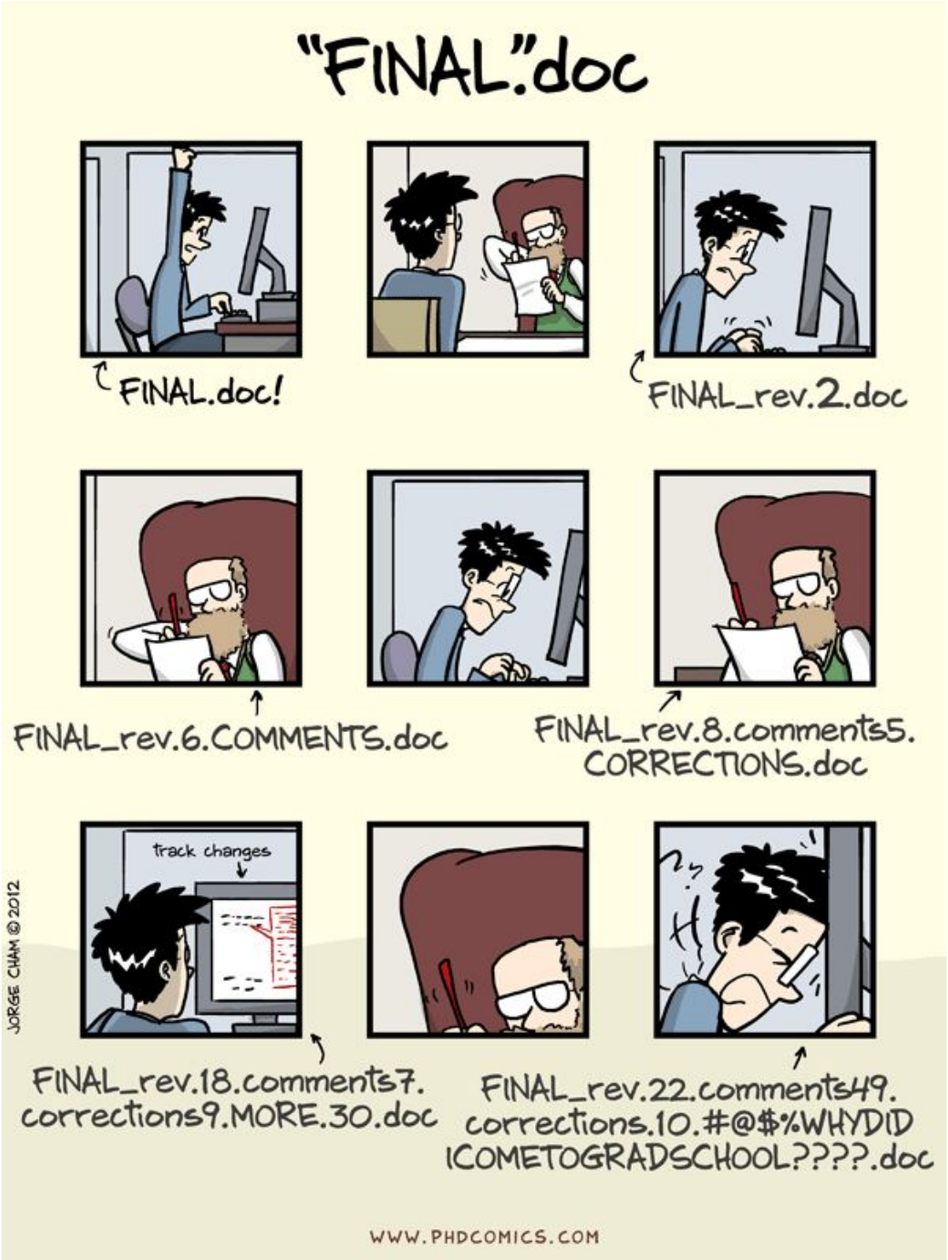
It is designed to be

- Simple and fast
- Fully distributed
- Able to support non-linear development
- Scalable

You won’t need all its features

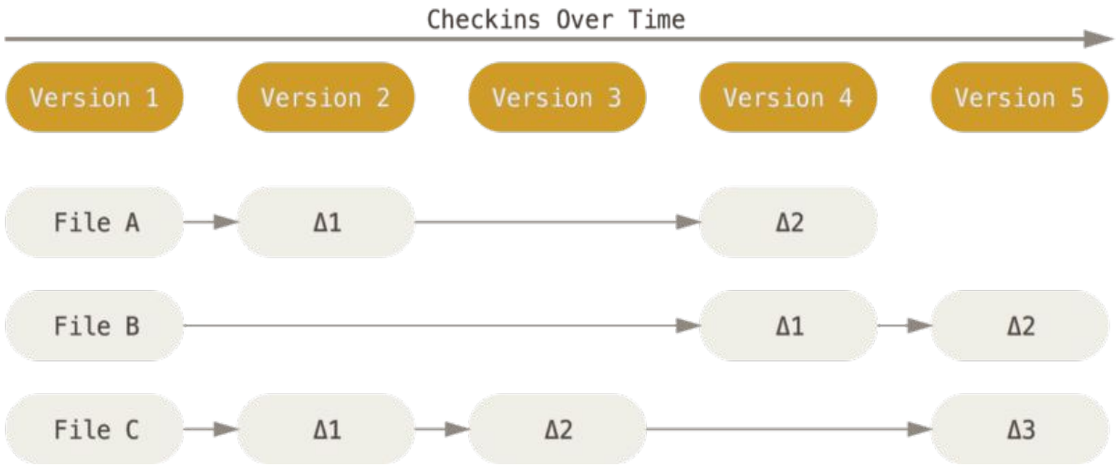


An intuitive user experience

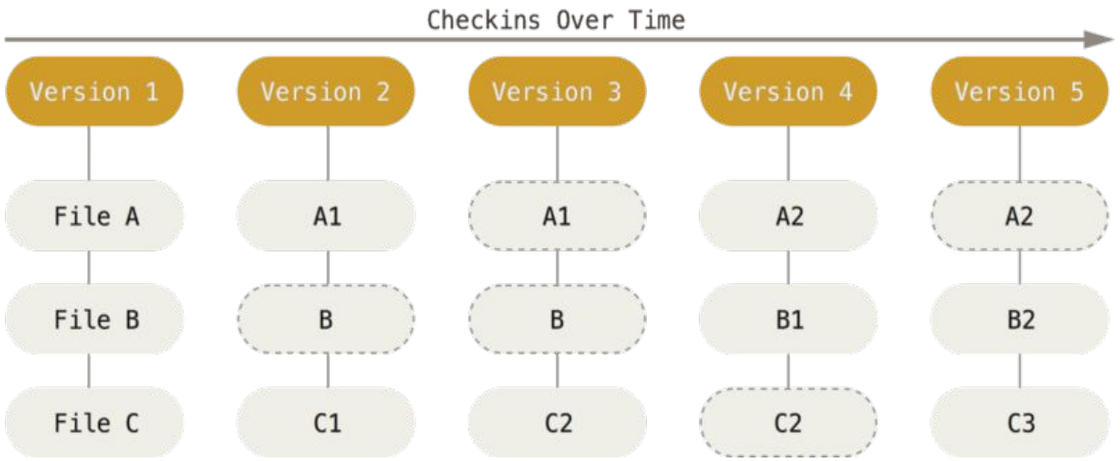


How does git do this?

“Delta-based version control” (other version control systems)



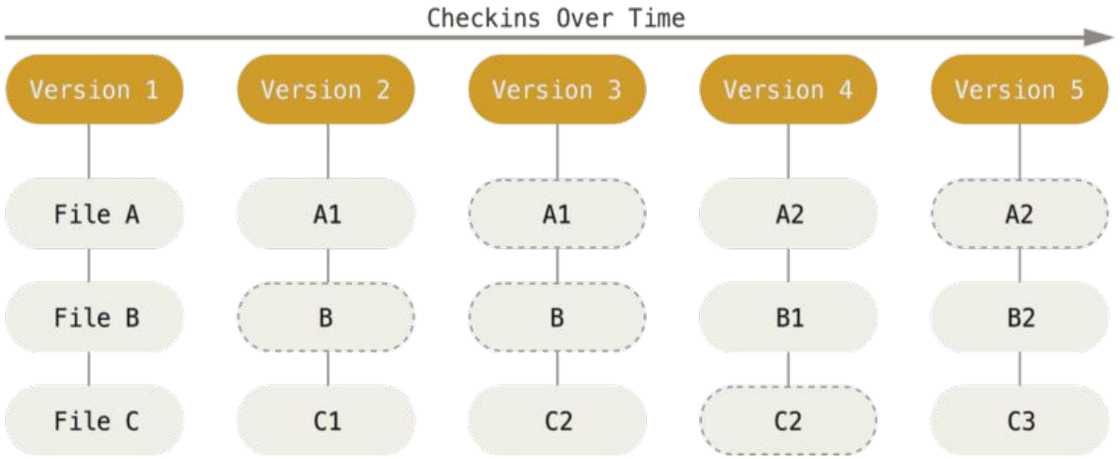
Stream of **snapshots** (git)



Git is reliable

- It doesn't rely on filenames to keep track of files

- It converts the contents of a file/directory → hash
  - E.g., `24b9da6552252987aa493b52f8696cd6d3b00373`
  - It is unique and deterministic (1-way)
  - Changed file content → changed hash



Goals

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- What is distributed version control?
- Why is Git useful?
- **Track your own work with Git**

Part 2 (next week)

- Share your work on GitHub (simple / linear collaboration)
- Collaborate through GitHub (complex / nonlinear collaboration)
- Navigating GitHub
- Try it out!

Check if you're ready

If you're **stuck** on a step, enter the step **letter** in the **comments** on Zoom

☑️ A) Can you open a bash shell?

- Open a terminal, type `echo $SHELL` and press ENTER.
- The output should be `/bin/bash`

☑️ B) Do you have git installed?

- In the bash terminal, `git --version` and press ENTER.
- The output should be `git version X` (where the X is the version number)
- *Don't worry if you don't have the exact same version as I do*

☑️ C) Do you have git configured?

- In the bash terminal, type `git config --list` and press ENTER
- You should see your **name** and **email**
  (and other things that aren't essential to configure)
- The email should be the one you used to sign up for **GitHub**

☑️ D) Can you open a text editor? E.g.,

- Linux: gedit, nano
- macOS: textedit
- Windows: notepad

Basic Bash

- `cd <directory path>` changes directories (aka folders)
  - `.` represents the current directory
  - `..` represents the directory one level up

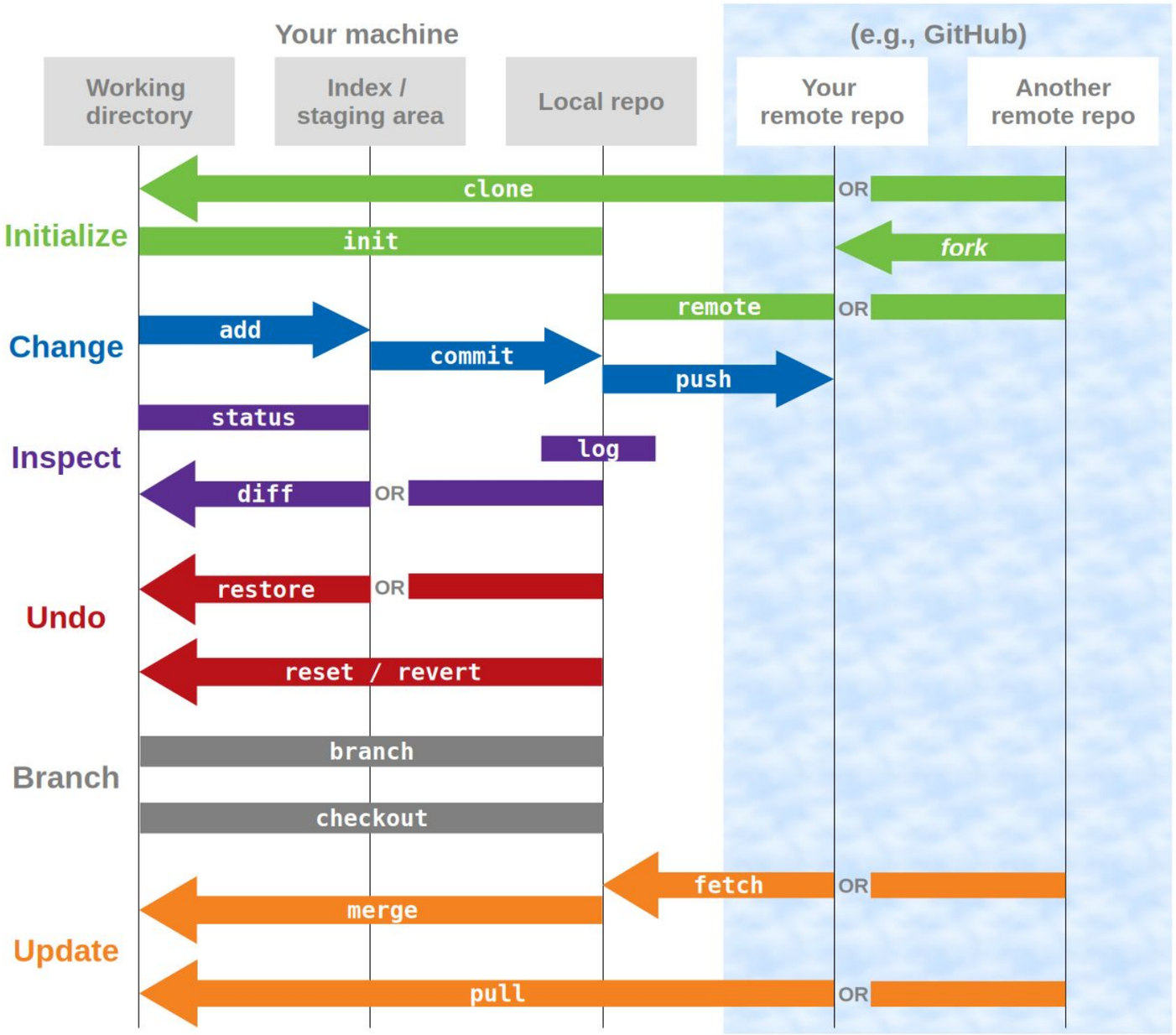
- `mkdir <directory path>` makes a new directory

- `touch <filename>` creates a new file

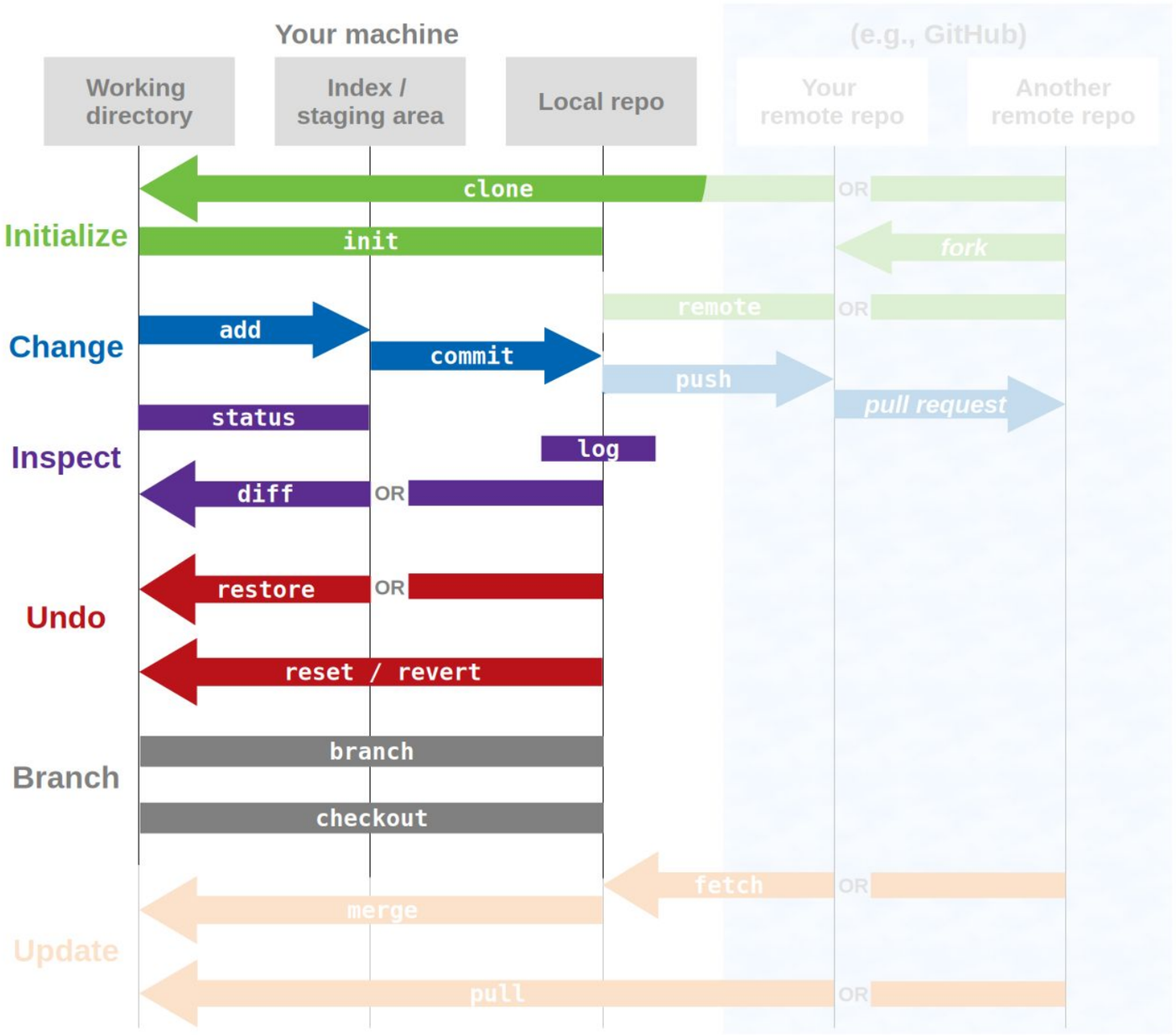
- `ls` lists the directory's contents
  - `ls -a` includes hidden files

- `<text editor name>` opens a text editor
  - `<text editor name> <filename>` opens a file in a text editor

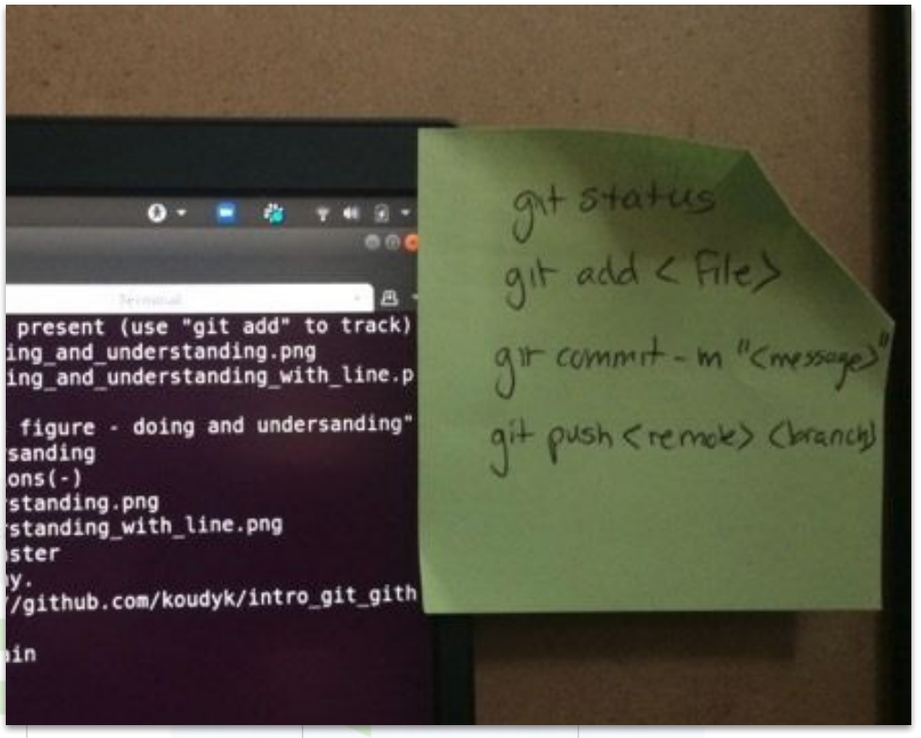
The commands we'll cover **in parts 1 and 2**

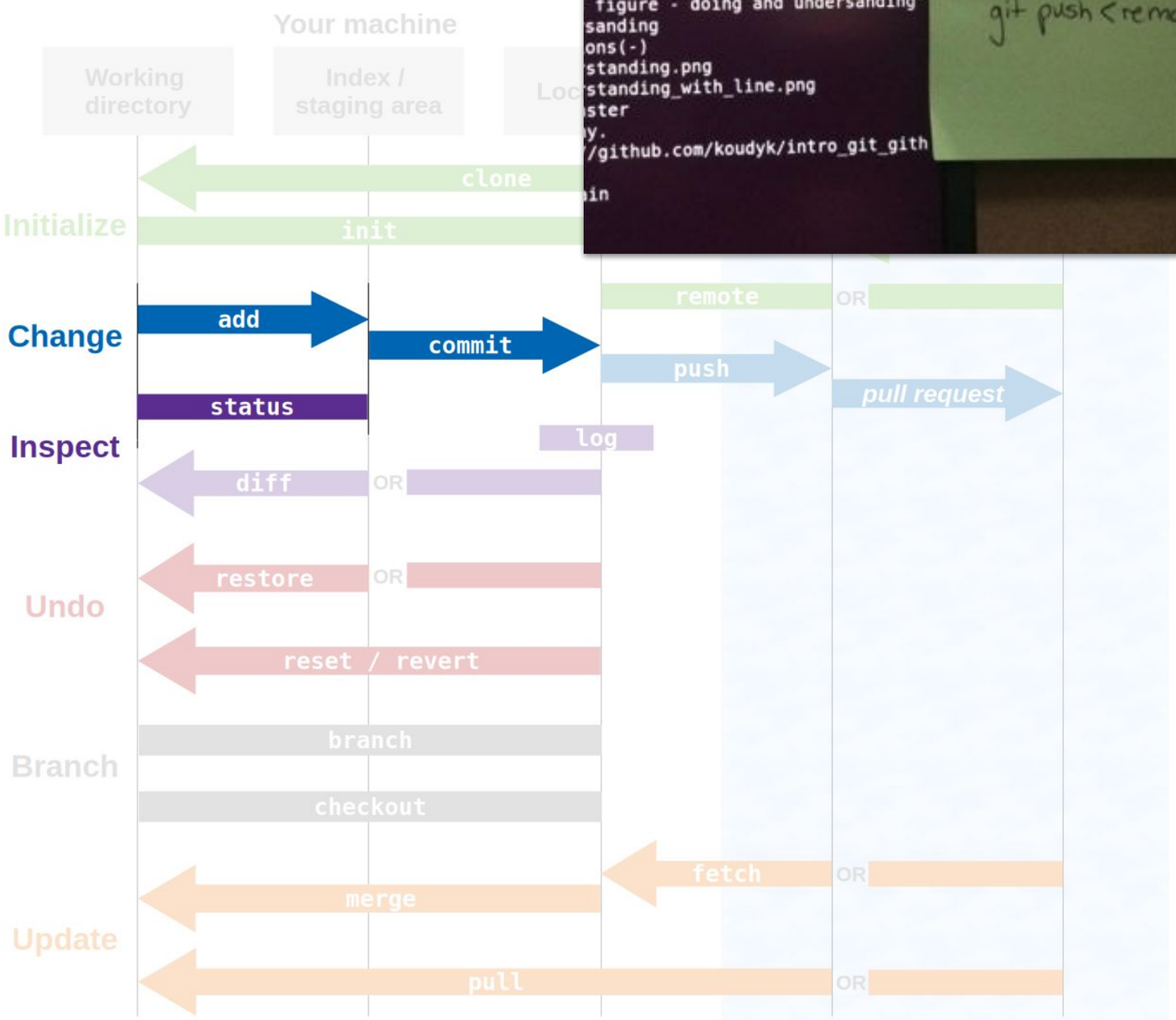


The commands we'll cover **in part 1**



The commands **you'll need to remember**



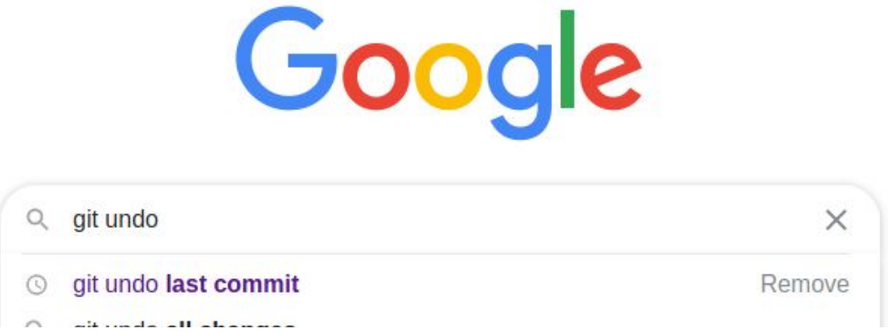


(Everything else you can look up when you need it)

First, how to get help

- In the terminal
  (if you know the verb and want to know what it does or what are its options)
  `git help <verb>`
  `git <verb> --help or git <verb> -h`
  `man git-<verb>`
  TIP: press 'q' to exit the manual in the terminal

- Look it up

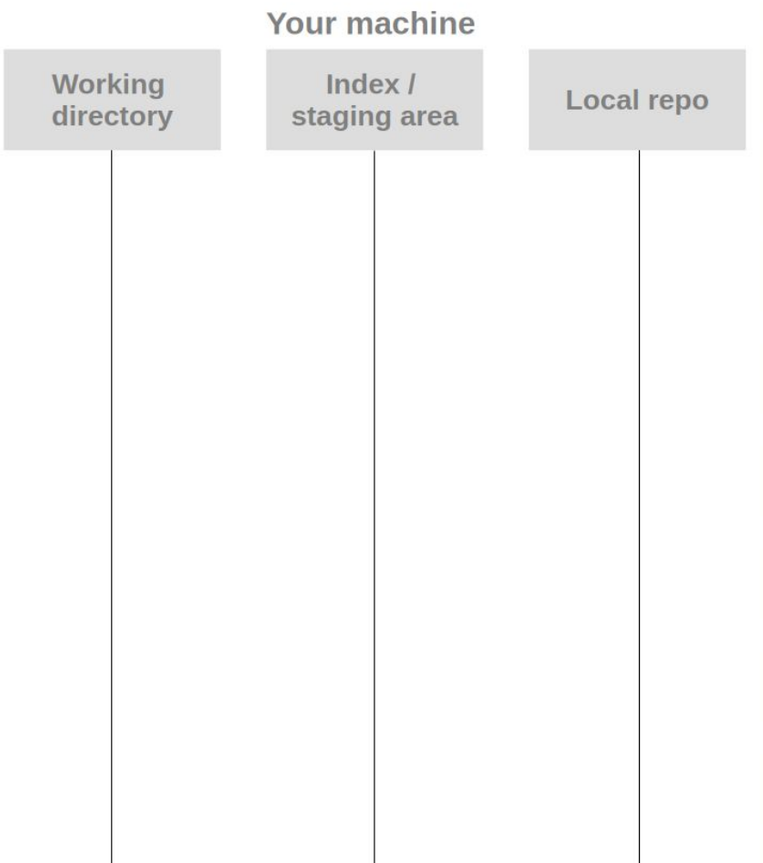


3 parts of a Git project

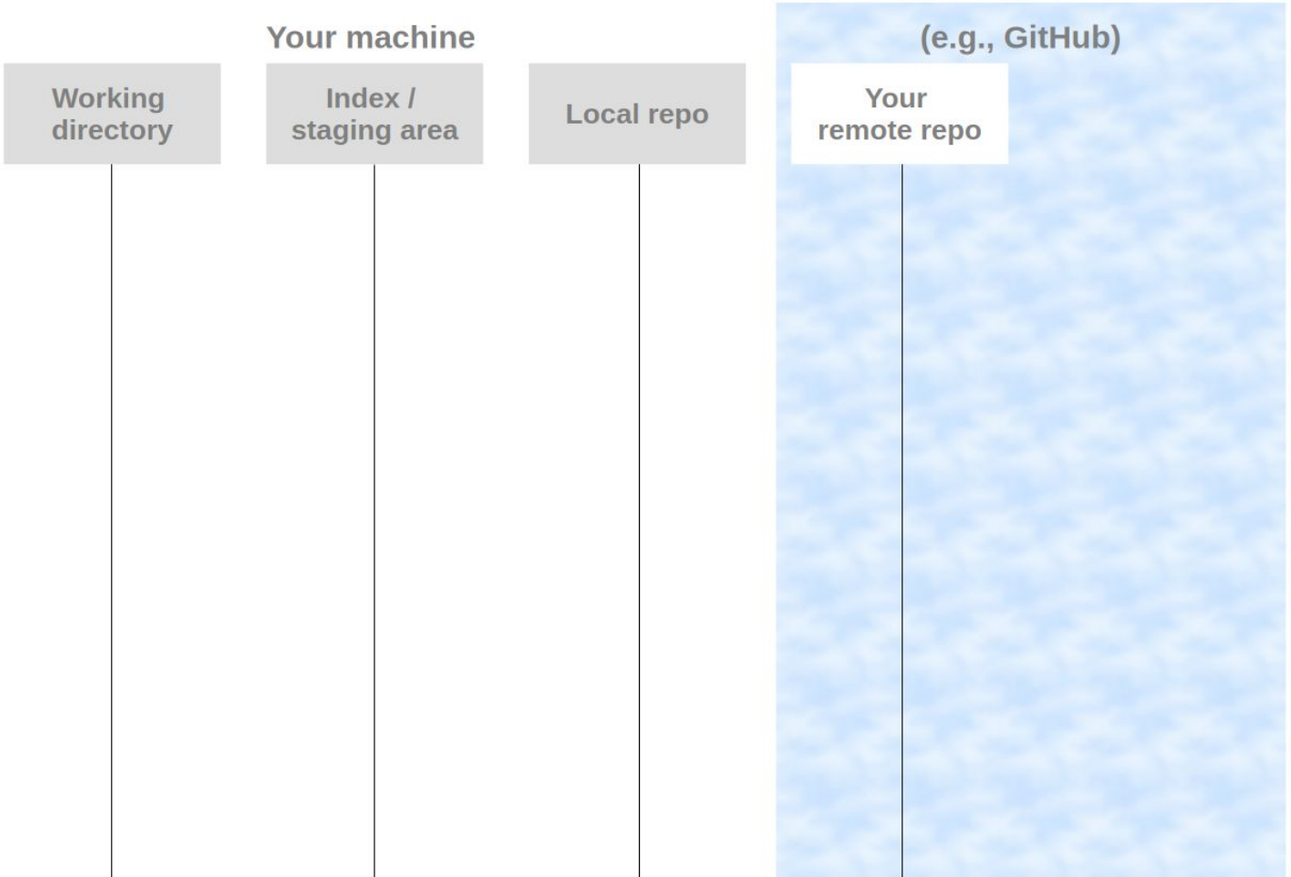
1. **Working directory**
   The version of the project that you're working on

2. **Staging area / Index**
   What will be in your next snapshot

3. **Local repository (i.e., `.git/` folder)**
   Metadata and objects that make up the snapshots



Peek forward: Where’s GitHub?



Starting a git repo

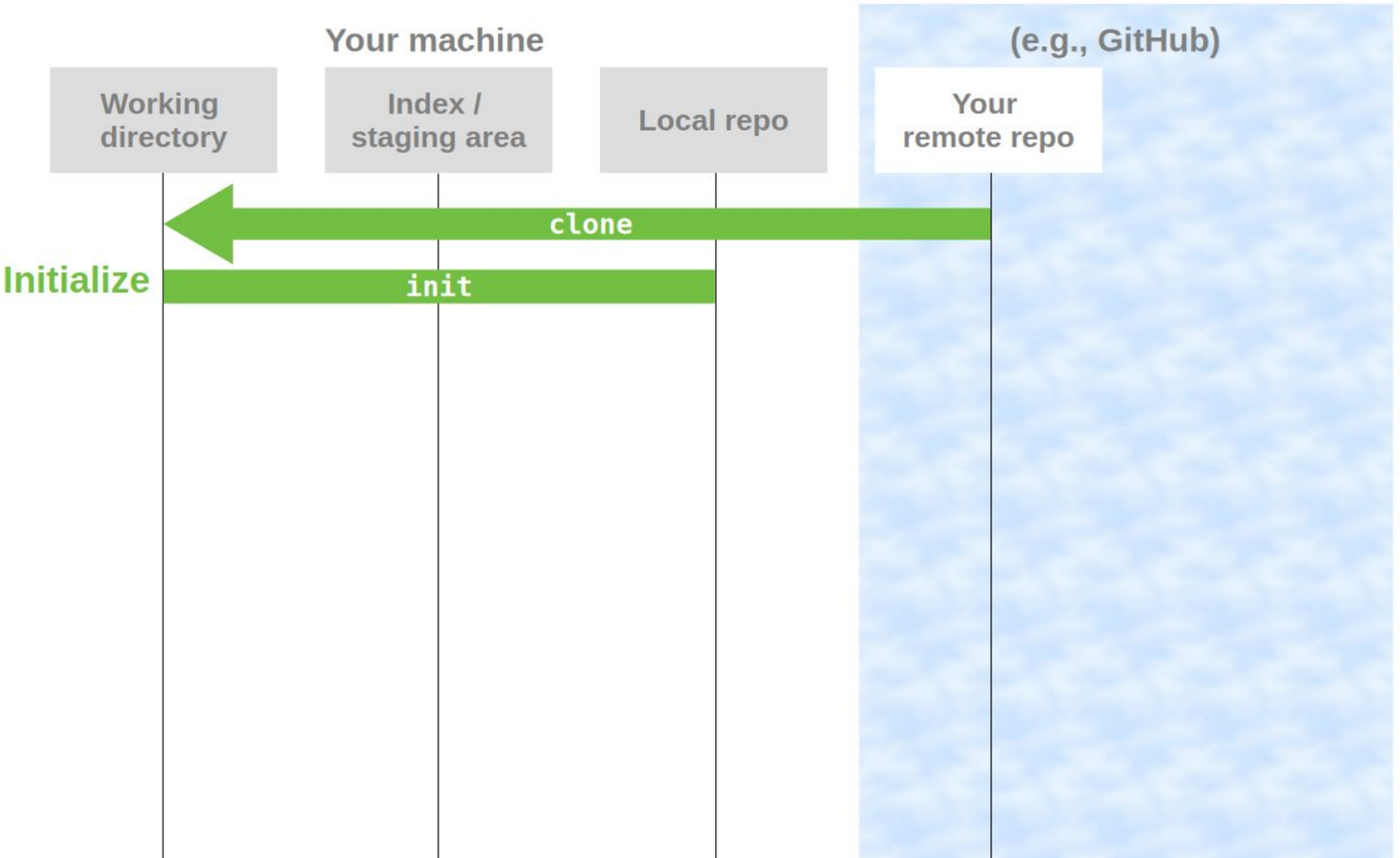
2 options

1. **Make an existing folder into a git repo**

   ```
   cd <directory>
   git init
   ```

2. **Clone an existing repo (e.g., from GitHub)**

   ```
   git clone <repo URL>
   ```



3 file states

1. **Modified**
   You made a change to the file

2. **Staged**
   You indicated that you want the modified file in your next snapshot

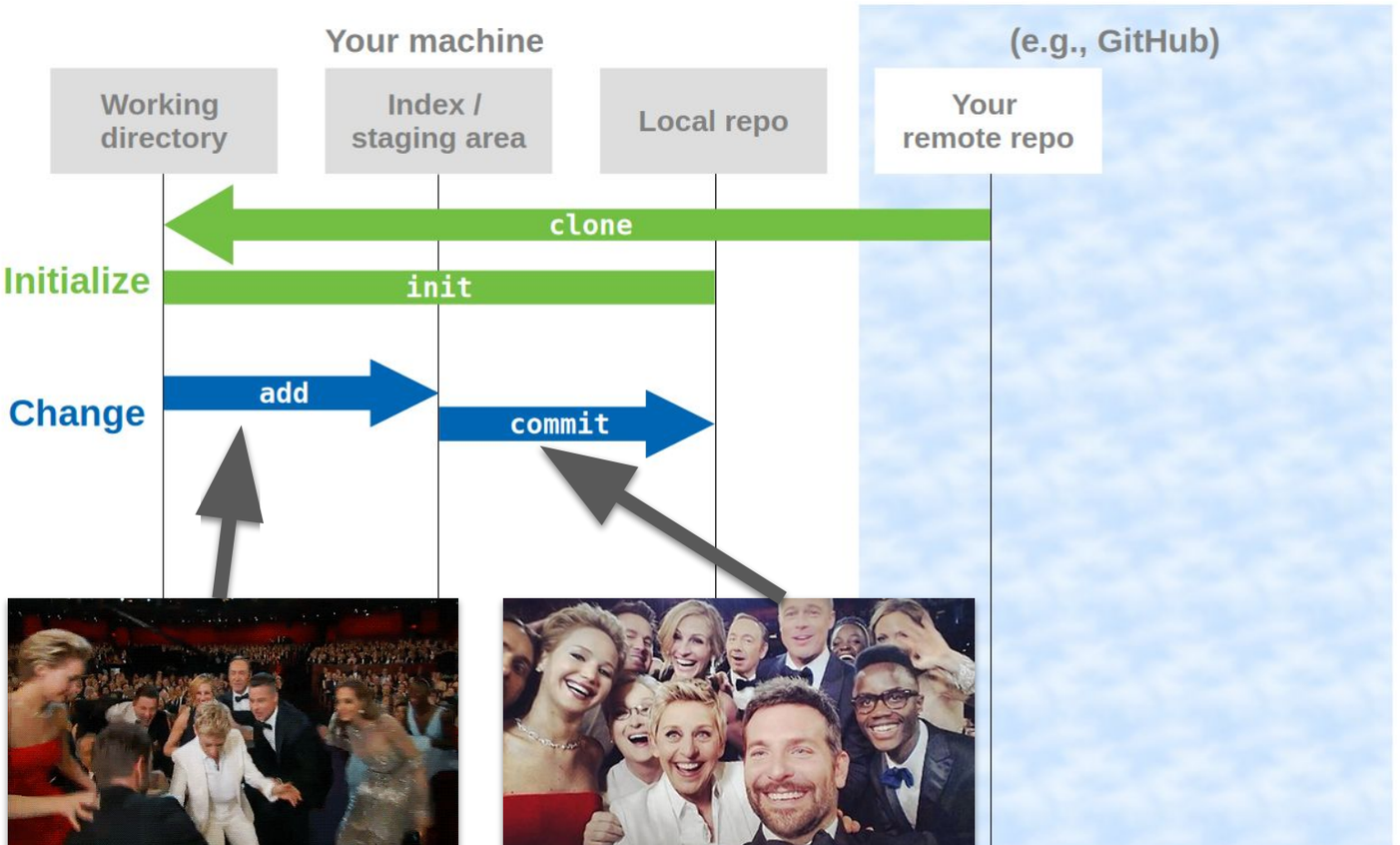
3. **Committed**
   You took the snapshot

3-step basic workflow

1. **Modify**
   Change a file in your working tree

2. **Stage**
   `git add <filename>`

3. **Commit**
   `git commit -m "<short, informative commit message>"`



Review: Tracking your work

Put these steps in order:

A) `git commit`

B) Change a file in your working tree

C) `git add`

Review: Tracking and sharing your work

Put these steps in order:

A) Take the snapshot
`git commit`

B) Change a file in your working tree

C) Get the altered file ready to be in the snapshot
`git add`

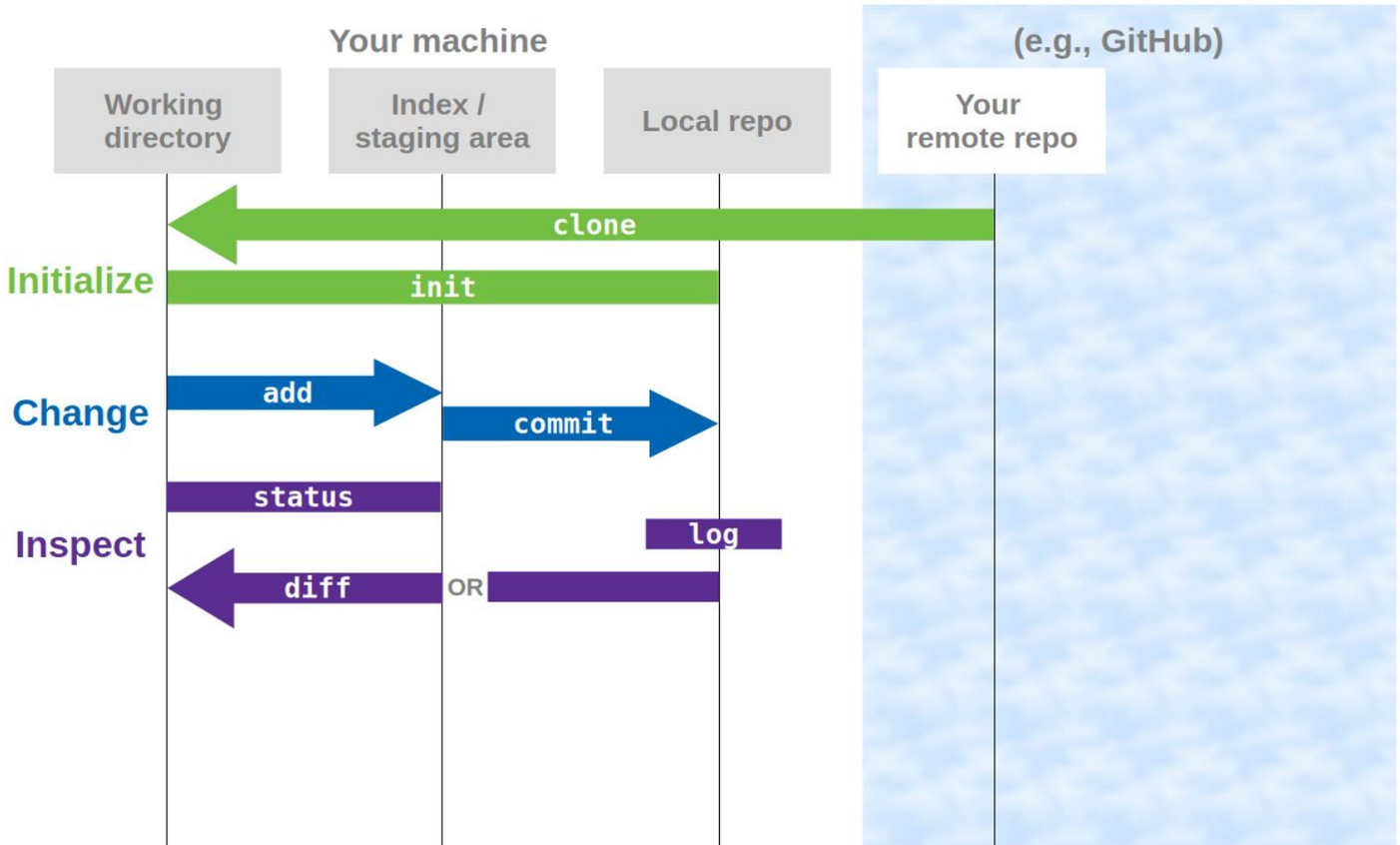
Inspecting

(useful commands that don't *do* anything)

- Check the status of the files in your repo
  `git status`

- See what changed
  `git diff`

- See the history of your repo
  `git log`
  Note: type 'q' to exit the log



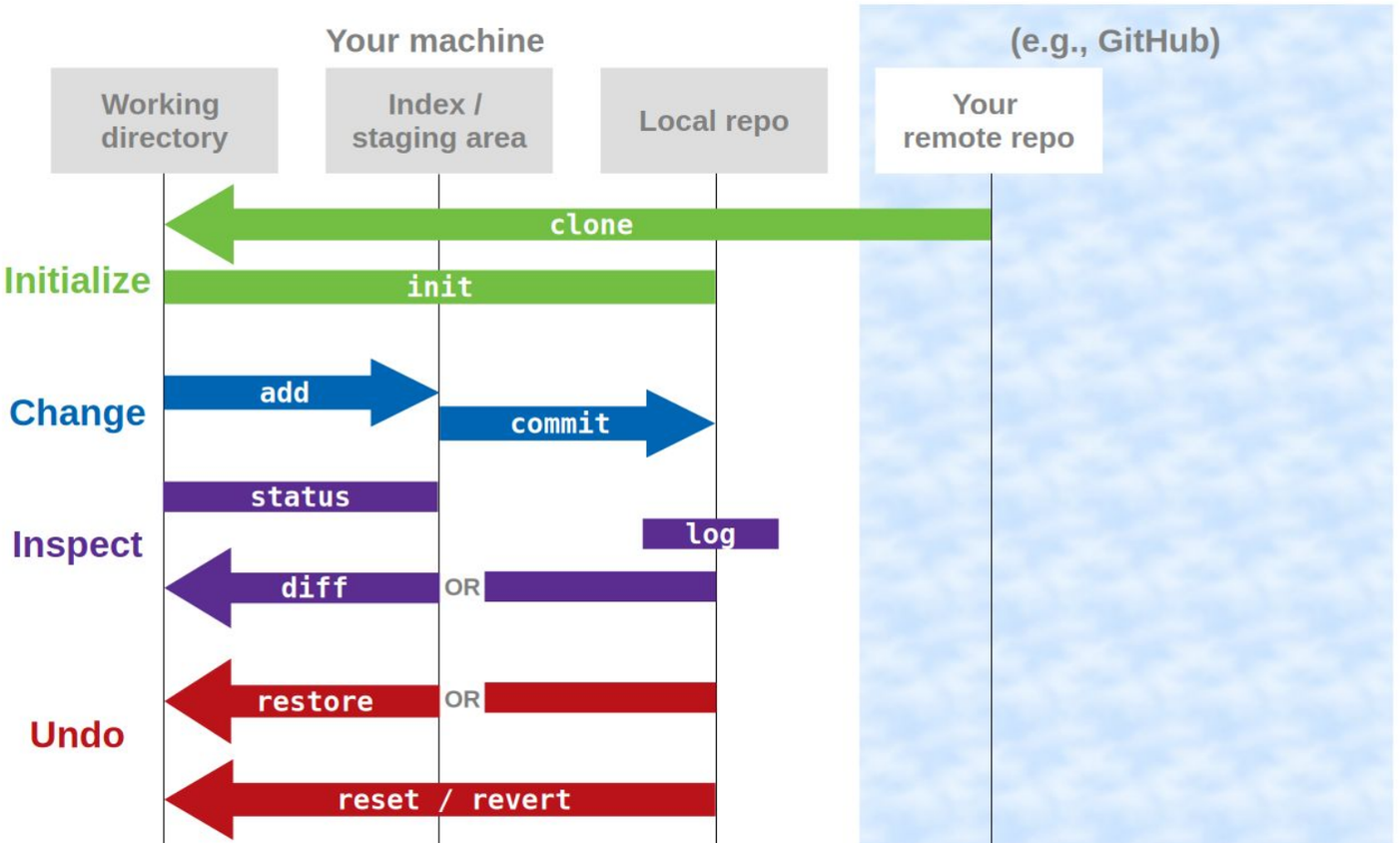
Common undoing goals

- **Unmodify** a file
  `git restore <file>` or `git checkout -- <file>`

- **Unstage** a file
  `git restore --staged <file>` or `git reset HEAD <file>`

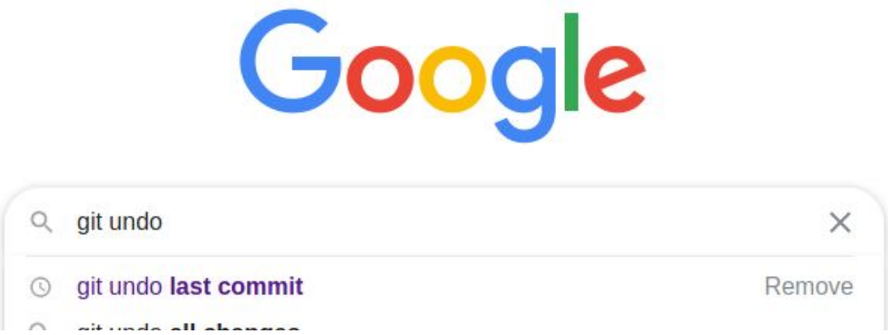
- Forgot a file in the last **commit**
  `git add <file>`
  `git commit --amend`

- Undo the last **commit**
  `git reset HEAD~`



Undoing can feel intimidating

- Don’t try to memorize all the commands
- Try it out on a sandbox repo first
- Use `git checkout <commit hash> -b <branch name>` to start a new branch at the point you want to revert to in order to see what it would be like
- For minor recent fixes, try reading the output of `git status`
- Look it up; someone probably had your question before



Review: Inspecting and changing

Which of these make a change?

A) git restore

B) git diff

C) git log

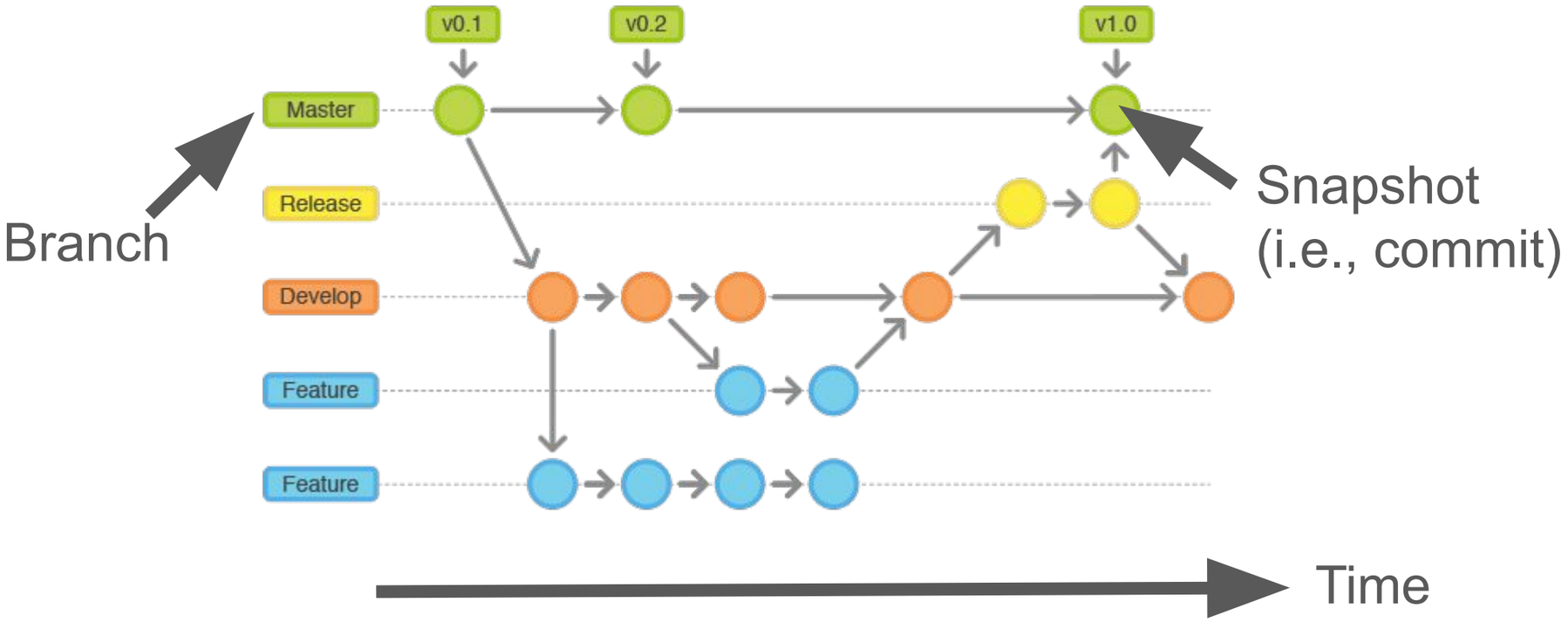
D) git status

E) git reset

Branches

- For nonlinear development
- To collaborate
  - Linux: In the last month, “1067 authors have pushed 5,329 commits…” [https://github.com/torvalds/linux/pulse/monthly, as of 2021-07-14]
- To try new things on your own

Branches



Tip: GitHub has a great way of viewing a project's "network"

Branches vs tags

	What is it for?	
Branch	**Marks a line of development** E..g, a new feature, a collaborator's contribution	
Tag	**Marks an important point in history** E..g, a version of a software package, a paper publication	

Branches vs tags

	What is it for?	What *exactly* is it?
Branch	**Marks a line of development** E..g, a new feature, a collaborator's contribution	**A text file** Filename: branch name Contents: commit hash for the latest commit in that branch
Tag	**Marks an important point in history** E..g, a version of a software package, a paper publication	**A text file** Filename: tag name Contents: commit hash for the commit when the tag was created

Branches

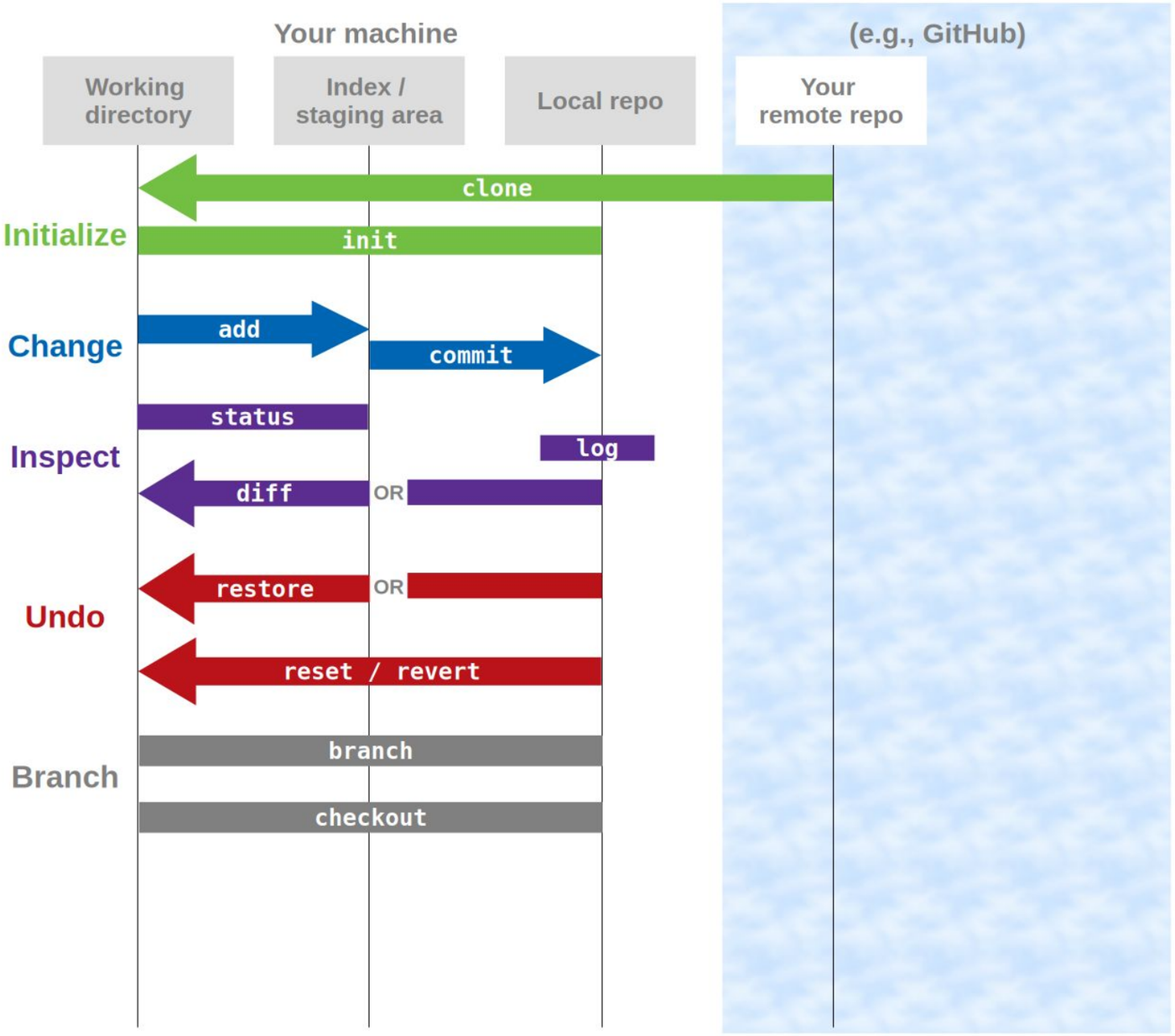
- See which branch you're on
  `git branch`

- Start a new branch
  `git branch <branch name>`

- Change branches
  `git checkout <branch name>`

- Merge a branch into your current branch
  `git merge <branch name>`

Branches

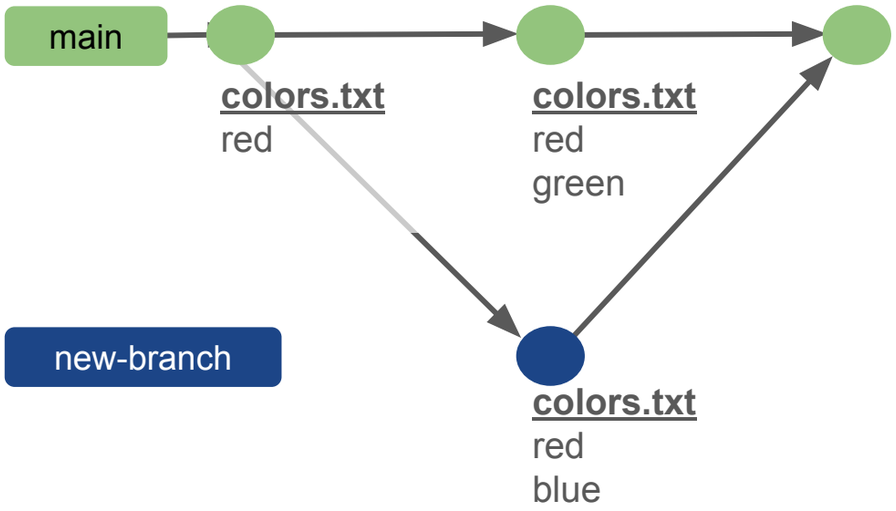

Your machine
(e.g., GitHub)
Working directory
Index / staging area
Local repo
Your remote repo
clone
Initialize
init
Change
add
commit
status
Inspect
log
diff
OR
restore
OR
Undo
reset / revert
branch
Branch
checkout

Merge conflicts

- What if **several people** edit the same line of code?
  (or **you** edit the same line of code on different branches)

  → **merge conflict**

- Someone needs to **manually resolve it**



colors.txt

```
red
Accept Current Change | Accept Incoming Change | Accept Both Changes | Compar
<<<<<<< HEAD (Current Change)
green
=======
blue
>>>>>>> new-branch (Incoming Change)
```

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Review of Part 1

How well could you explain the following to a peer?

(use the Zoom annotate tool)

A) **What is distributed version control?**

With difficulty---------- Easily

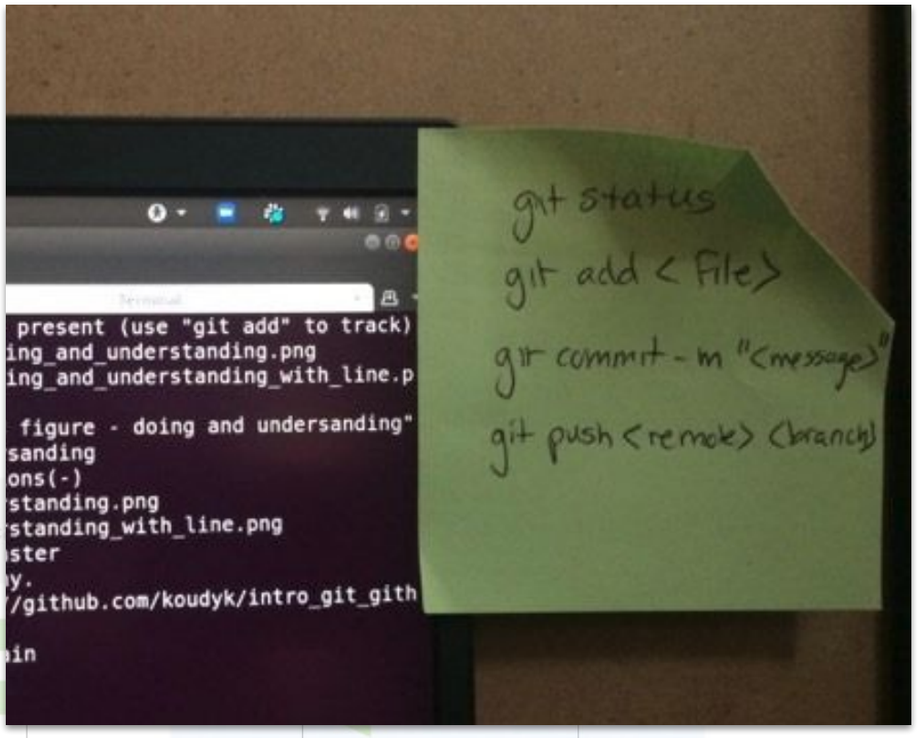
B) **Why is Git useful?**

With difficulty---------- Easily

C) **How do I track my own work with Git?**

With difficulty---------- Easily

The commands **you'll need to remember**



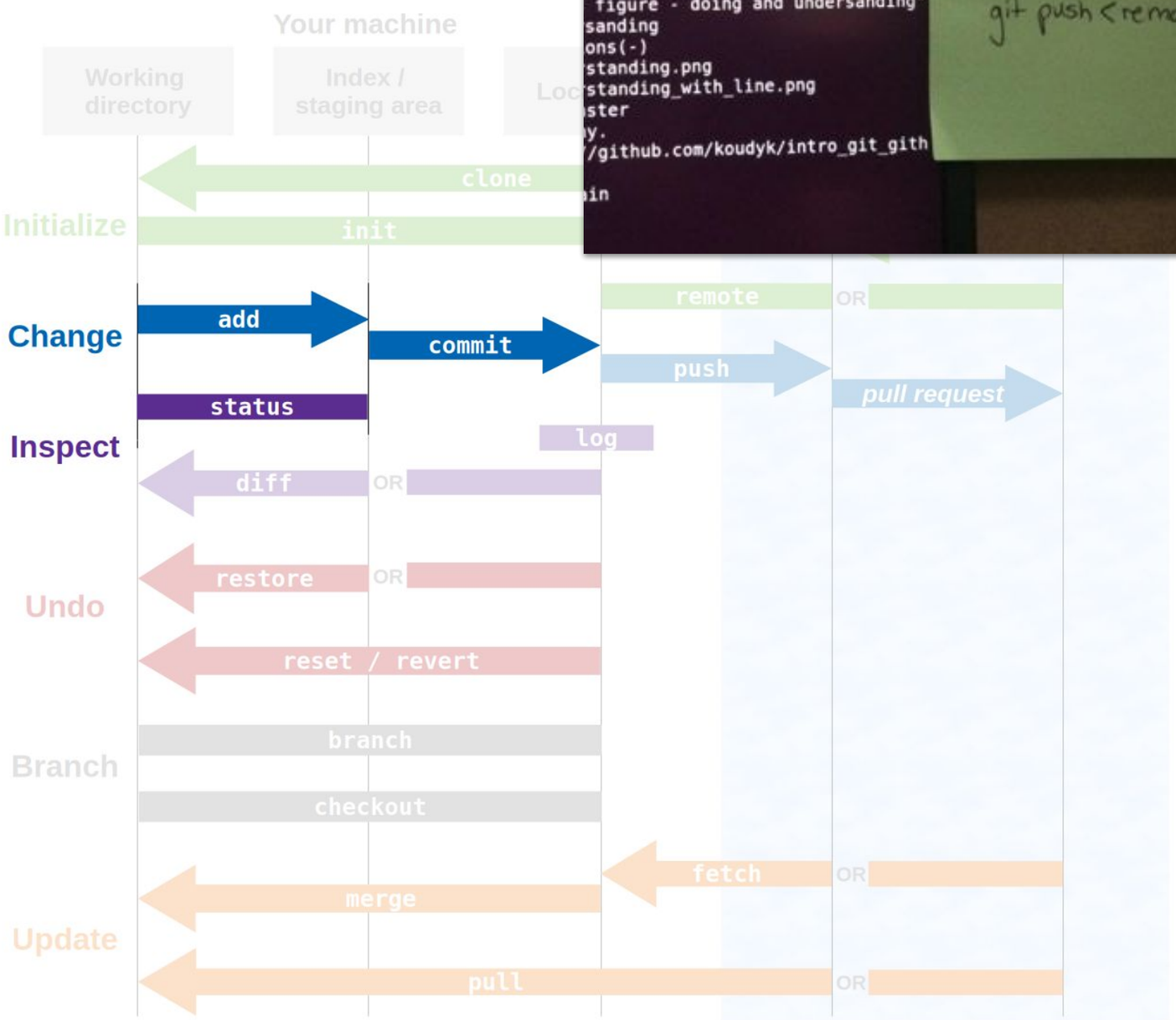


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Figure references

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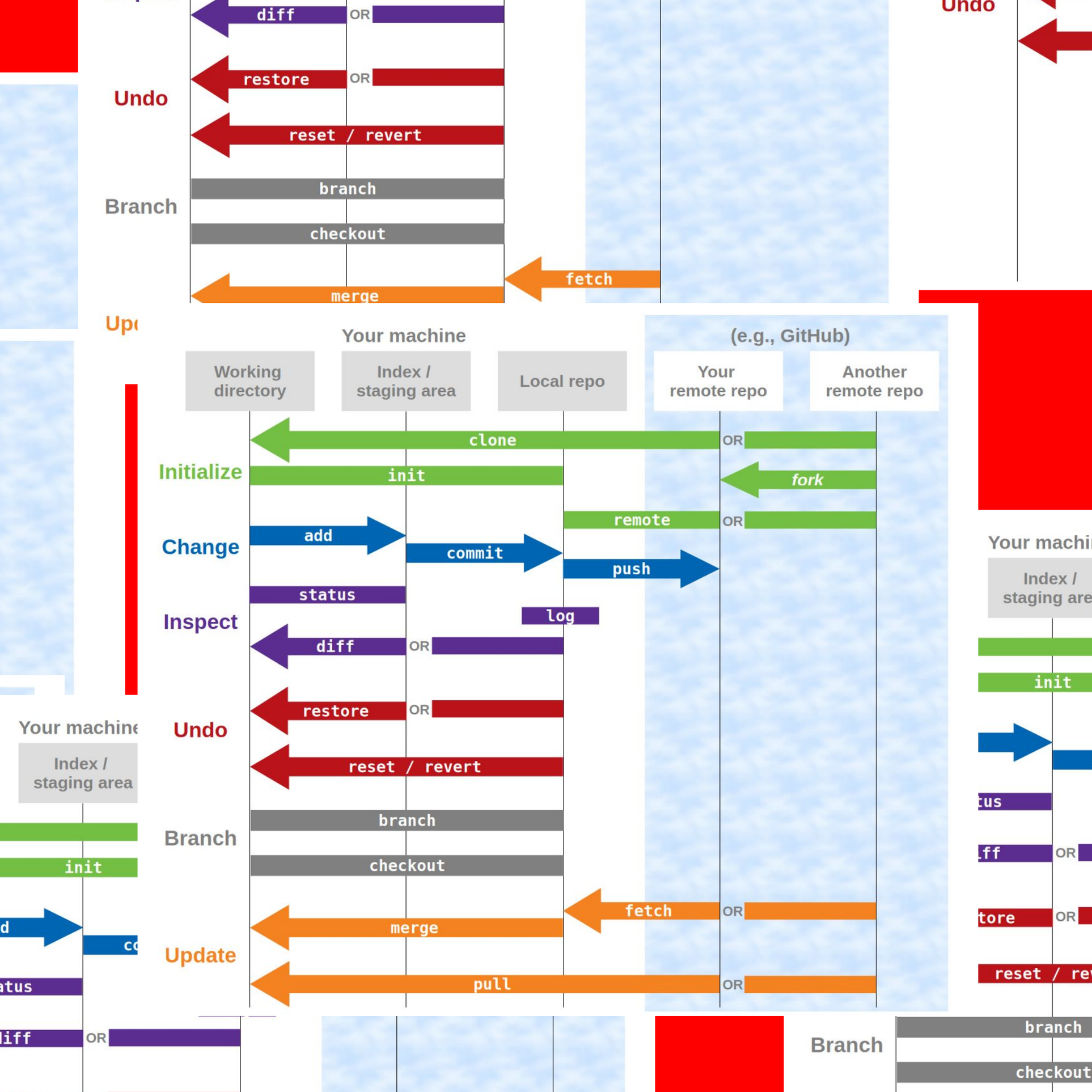
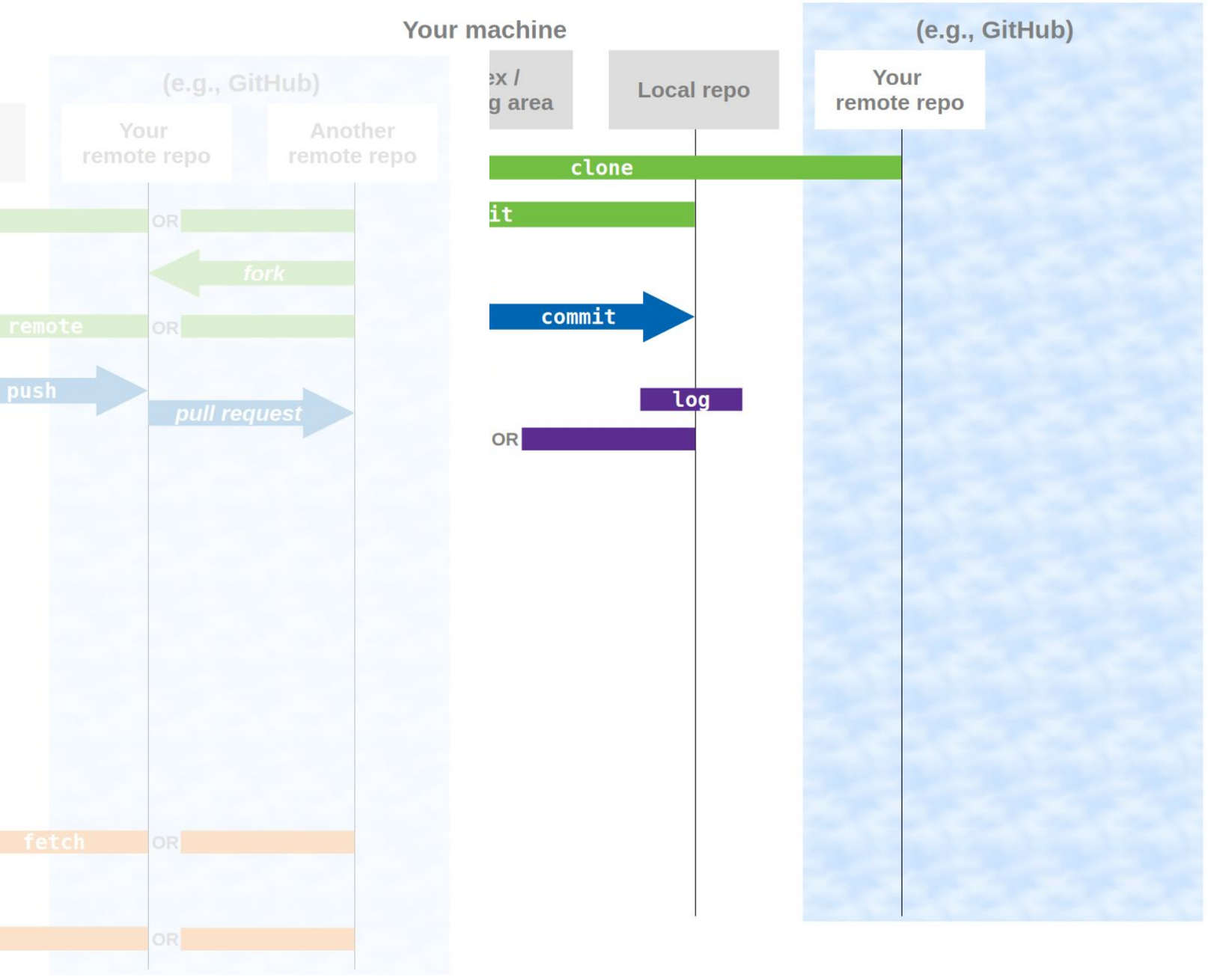
- www.phdcomics.com 15 - https://en.wikipedia.org/wiki/Git
- McElreath, R. (2020, September 26). Science as amateur software development [video]. YouTube. • Science as Amateur Software Development

The end


Your machine
(e.g., GitHub)
Working directory
Index / staging area
Local repo
Your remote repo
Another remote repo
Initialize
clone
OR
init
fork
Change
remote
OR
add
commit
push
Inspect
status
log
diff
OR
Undo
restore
OR
reset / revert
Branch
branch
checkout
Update
fetch
OR
merge
pull
OR


Your machine
(e.g., GitHub)
(e.g., GitHub)
Local repo
Your remote repo
Your remote repo
Another remote repo
clone
it
OR
fork
commit
remote
OR
push
log
pull request
OR
fetch
OR
OR